大学物理实验报告

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实验三十一　冷却法测量金属的比热容

［目的］

［原理］

［数据记录与分析］

1．试样质量和热电偶冷端温度：　T_0=_____ ℃

表 1 试样质量（g）

样品＼次数	1	2	3	4	5	平均值 $\overline{m}$
Cu						
Fe						
Al						

$u_0(\overline{m_{Cu}}) = S_0(\overline{m_{Cu}}) = \sqrt{\frac{\sum(\Delta m_i)^2}{N(N-1)}}$　　$u_1(\overline{m_{Cu}}) = \frac{0.05}{\sqrt{3}} =$　　g

$u_{cm_{Cu}} = \sqrt{u_0(\overline{m_{Cu}})^2 + u_1(\overline{m_{Cu}})^2}$

$u_0(\overline{m_{Fe}}) = S_0(\overline{m_{Fe}}) = \sqrt{\frac{\sum(\Delta m_i)^2}{N(N-1)}}$　　$u_1(\overline{m_{Fe}}) = \frac{0.05}{\sqrt{3}} =$　　g

$u_{m_{Fe}} = \sqrt{u_0(\overline{m_{Fe}})^2 + u_1(\overline{m_{Fe}})^2}$

$u_0(\overline{m_{Al}}) = S_0(\overline{m_{Al}}) = \sqrt{\frac{\sum(\Delta m_i)^2}{N(N-1)}}$　　$u_1(\overline{m_{Al}}) = \frac{0.05}{\sqrt{3}} =$　　g

$u_{m_{Al}} = \sqrt{u_0(\overline{m_{Al}})^2 + u_1(\overline{m_{Al}})^2}$

表 2 特定温度对应的热电势（mV）

$E_{(150℃,\ 0℃)}$	$E_{(102℃,\ 0℃)}$	$E_{(98℃,\ 0℃)}$	$E_{(t0,\ 0℃)}$	$E_{(150℃,\ t0)}$	$E_{(102℃,\ t0)}$	$E_{(102℃,\ t0)}$

2．样品热电势由 $E_1=E_{(102℃,\ t0)}=E_{(102℃,\ 0℃)}-E_{(t0,\ 0℃)}$ 下降到 $E_2=E_{(98℃,\ t0)}=E_{(98℃,\ 0℃)}-E_{(t0,\ 0℃)}$ 所需时间（单位为 S/秒）

表 2 冷却时间(s)

样品＼次数	1	2	3	4	5	平均值 $\overline{\triangle t}$
Cu						
Fe						
Al						

以铜为标准：　C_{cu}=0.0940 cal/（g K）

$u_0(\Delta\overline{t_{Cu}}) = S_0(\Delta\overline{t_{Cu}}) = \sqrt{\frac{\sum(\Delta t_i)^2}{N(N-1)}}$

计时器分辨力引入 $u_1(\Delta\overline{t_{Cu}}) = \frac{0.005}{\sqrt{3}} =$ s 计时器内部误差 $u_2(\Delta\overline{t_{Cu}}) = \frac{0.01}{\sqrt{3}} =$ s

$u\Delta_{t_{Cu}} = \sqrt{u_0(\Delta\overline{t_{Cu}})^2 + u_1(\Delta\overline{t_{Cu}})^2 + u_2(\Delta\overline{t_{Cu}})^2} =$

$u_0(\Delta\overline{t_{Fe}}) = S_0(\Delta\overline{t_{Fe}}) = \sqrt{\frac{\sum(\Delta t_i)^2}{N(N-1)}}$

$u_1(\Delta\overline{t_{Fe}}) = \frac{0.005}{\sqrt{3}} =$ s $u_2(\Delta\overline{t_{Fe}}) = \frac{0.01}{\sqrt{3}} =$ s

$u\Delta_{t_{Fe}} = \sqrt{u_0(\Delta\overline{t_{Fe}})^2 + u_1(\Delta\overline{t_{Fe}})^2 + u_2(\Delta\overline{t_{Fe}})^2} =$

铁的比热容： $c_{Fe} = c_{Cu}\frac{m_{Cu}\times \Delta t_{Fe}}{m_{Fe}\times \Delta t_{Cu}} =$ Cal/（g K）

铁的比热容的标准不确定度

$$u_{c_{Fe}} = \sqrt{(\frac{\partial f}{\partial m_{Cu}}u_{M_{Cu}})^2 + (\frac{\partial f}{\partial \Delta t_{Fe}}u_{\Delta t_{Fe}})^2 + (\frac{\partial f}{\partial m_{Fe}}u_{M_{Fe}})^2 + (\frac{\partial f}{\partial \Delta t_{Cu}}u_{\Delta t_{Cu}})^2}$$

求铁的比热容的扩展不确定度

各分量的有效自由度

$$v_{eff}(m_{Cu}) = \frac{u_{m_{Cu}}^4}{\frac{u_0(\overline{m_{Cu}})^4}{5-1} + \frac{u_1(\overline{m_{Cu}})^4}{\infty}}$$

$$v_{eff}(\Delta t_{Cu}) = \frac{u_{\Delta t_{Cu}}^4}{\frac{u_0(\overline{\Delta t_{Cu}})^4}{5-1} + \frac{u_1(\overline{\Delta t_{Cu}})^4}{\infty}}$$

$$v_{eff}(m_{Fe}) = \frac{u_{m_{Fe}}^4}{\frac{u_0(\overline{m_{Fe}})^4}{5-1} + \frac{u_1(\overline{m_{Fe}})^4}{\infty}}$$

$$v_{eff}(\Delta t_{Fe}) = \frac{u_{\Delta t_{Fe}}^4}{\frac{u_0(\overline{\Delta t_{Fe}})^4}{5-1} + \frac{u_1(\overline{\Delta t_{Fe}})^4}{\infty}}$$

c_{Fe}的有效自由度

$$v_{eff}(c_{Fe}) = \frac{u_{c_{Fe}}^4}{\frac{(\frac{\partial f}{\partial m_{Cu}}u_{M_{Cu}})^4}{v_{eff}(m_{Cu})} + \frac{(\frac{\partial f}{\partial \Delta t_{Fe}}u_{\Delta t_{Fe}})^4}{v_{eff}(\Delta t_{Fe})} + \frac{(\frac{\partial f}{\partial m_{Fe}}u_{M_{Fe}})^4}{v_{eff}(m_{Fe})} + \frac{(\frac{\partial f}{\partial \Delta t_{Cu}}u_{\Delta t_{Cu}})^4}{v_{eff}(\Delta t_{Cu})}}$$

则扩展不确定度 $U_{0.95}$=$k_{0.95}$ $\times u_{c_{Fe}}$=

$c_{Fe} =$, P=0.95

$u_0(\Delta\overline{t_{Al}}) = S_0(\Delta\overline{t_{Al}}) = \sqrt{\frac{\sum(\Delta t_i)^2}{N(N-1)}}$

$u_1(\Delta\overline{t_{Al}}) = \frac{0.005}{\sqrt{3}} =$ s $u_2(\Delta\overline{t_{Al}}) = \frac{0.01}{\sqrt{3}} =$ s

$u\Delta_{t_{Al}} = \sqrt{u_0(\Delta\overline{t_{Al}})^2 + u_1(\Delta\overline{t_{Al}})^2 + u_2(\Delta\overline{t_{Al}})^2} =$

$v_{eff}(t_{Al}) = \frac{u_{t_{Al}}^4}{\frac{u_0(\overline{t_{Al}})^4}{5-1} + \frac{u_1(\overline{t_{Al}})^4}{\infty}}$

铝的比热容： $c_{Al} = c_{Cu}\frac{M_{Cu}\ (\Delta t)_{Al}}{M_{Al}\ (\Delta t)_{Cu}} =$ Cal/（g K）

铝的比热容的标准不确定度

$$u_{c_{Fe}} = \sqrt{(\frac{\partial f}{\partial m_{Cu}}u_{M_{Cu}})^2 + (\frac{\partial f}{\partial \Delta t_{Al}}u_{\Delta t_{Al}})^2 + (\frac{\partial f}{\partial m_{Al}}u_{M_{Al}})^2 + (\frac{\partial f}{\partial \Delta t_{Cu}}u_{\Delta t_{Cu}})^2}$$

求铝的比热容的扩展不确定度

各分量的有效自由度

$$v_{eff}(m_{Fe}) = \frac{u_{m_{Fe}}^4}{\frac{u_0(\overline{m_{Fe}})^4}{5-1} + \frac{u_1(\overline{m_{Fe}})^4}{\infty}}$$

$$v_{eff}(\Delta t_{Fe}) = \frac{u_{\Delta t_{Fe}}^4}{\frac{u_0(\overline{\Delta t_{Fe}})^4}{5-1} + \frac{u_1(\overline{\Delta t_{Fe}})^4}{\infty}}$$

c_{Al}的有效自由度

$$v_{eff}(c_{Al}) = \frac{u_{c_{Al}}^4}{\frac{(\frac{\partial f}{\partial m_{Cu}}u_{M_{Cu}})^4}{v_{eff}(m_{Cu})} + \frac{(\frac{\partial f}{\partial \Delta t_{Fe}}u_{\Delta t_{Al}})^4}{v_{eff}(\Delta t_{Al})} + \frac{(\frac{\partial f}{\partial m_{Al}}u_{M_{Al}})^4}{v_{eff}(m_{Al})} + \frac{(\frac{\partial f}{\partial \Delta t_{Cu}}u_{\Delta t_{Cu}})^4}{v_{eff}(\Delta t_{Cu})}}$$

则扩展不确定度 $U_{0.95}$=$k_{0.95}$ $\times u_{c_{Al}}$=

$c_{Al} =$, P=0.95

[思考题]

1. 为什么实验在防风筒中进行？

2. 如果金属表面不光滑，或者表面有油脂污染，会对测量结果产生影响吗？请说明理由。